

Customer Shopping Behaviour Analysis

1. Project Overview

This project looks at how customers shop, what they purchase, how much they spend, and how their behaviour differs across customer groups.

The analysis combines Python, PostgreSQL and Power BI. Python was used to clean and prepare the data, PostgreSQL was used to answer business questions through SQL, and Power BI was used to turn the findings into an interactive dashboard.

The dataset contains 3,900 purchase records covering different product categories, customer demographics and shopping behaviours.

2. Data Preparation

The analysis started in Python using pandas.

The dataset was first inspected to understand its structure, data types and summary statistics. Missing values were then checked, with missing review ratings handled using the median rating within each product category.

The column names were standardized into a cleaner snake case format, which made them easier to work with in both Python and PostgreSQL.

A few additional features were created during the analysis:

- `age_group` to make age-based comparisons easier
- `purchase_frequency_days` to convert purchase frequency descriptions into numerical day values

A data consistency check was also performed on `discount_applied` and `promo_code_used`. Since the `promo_code_used` information was considered redundant for the analysis, the column was removed.

The cleaned DataFrame was then loaded into PostgreSQL for further analysis.

3. Key Business Findings

a) Revenue by Gender

Male customers generated \$157,890 in revenue, compared with \$75,191 from female customers.

That means male customers contributed roughly 68% of total revenue in the dataset.

	gender text 🔒	revenue numeric 🔒
1	Female	75191
2	Male	157890

b) High Spending Discount Users

The analysis identified 839 customers made a purchase where a discount was applied while the purchase amount was still above the overall average.

This is an interesting finding because it suggests that discounts are not necessarily limited to low value purchases.

	customer_id bigint 🔒	purchase_amount bigint 🔒
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60
18	41	76
19	43	100
20	44	69
21	55	94
...	...	...
Total rows: 839		Query complete 00:00:00.142

c) Top Rated Products

The five products with the highest average review ratings were Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), and Skirt (3.78). Gloves had the highest average rating.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

d) Shipping Type

The average purchase amount was \$60.48 for Express shipping compared with \$58.46 for Standard shipping. Express purchases were therefore slightly higher on average.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

e) Subscription Behaviour

There were 1,053 subscribers (27%) and 2,847 non-subscribers (73%). Average spending was almost identical at \$59.49 for subscribers and \$59.87 for non-subscribers.

This suggests that the main opportunity is increasing subscription adoption, rather than increasing spending among existing subscribers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	No	2847	59.87	170436.00
2	Yes	1053	59.49	62645.00

f) Products and Discounts

The products with the highest discount rates were Hat (50.00%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%).

Hat had the highest discount rate at 50%.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

g) Customer Loyalty

The customer segments consisted of 3,116 Loyal, 701 Returning, and 83 New customers. The large Loyal segment shows that repeat purchasing is a strong characteristic of the dataset.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

h) Product Preferences by Category

The most purchased products included Jewellery, Sunglasses and Belt in Accessories; Blouse, Pants and Shirt in Clothing; and Sandals, Shoes and Sneakers in Footwear.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

i) Repeat Buyers and Subscriptions

Among customers with more than five previous purchases, 2,518 were non-subscribers and 958 were subscribers.

This indicates an opportunity to target repeat buyers with subscription offers

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

k) Revenue by Age Group

Revenue was relatively balanced across age groups. Young Adults generated the highest revenue at \$62,143, followed by Middle aged (\$59,197), Adult (\$55,978) and Senior (\$55,763).

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

4. What the Power BI Dashboard Adds

The Power BI dashboard brings these findings together into one interactive view.



5. Business Recommendations

A. Increase Subscription Adoption

Encourage regular customers to subscribe through exclusive discounts, faster shipping and loyalty benefits.

B. Target Repeat Buyers

Promote subscriptions to repeat buyers who already show strong engagement.

C. Review Discount Strategy

Monitor highly discounted products to ensure discounts increase sales without hurting margins.

D. Prioritize Key Categories

Continue focusing on Clothing and Accessories, which generate most of the revenue.

E. Promote Top Rated Products

Highlight highly rated products such as Gloves, Sandals and Boots in marketing campaigns.

F. Strengthen Customer Acquisition

Continue attracting new customers to balance the large Loyal customer segment.

6. Final Conclusion

Overall, the analysis shows a customer base with strong repeat purchase behaviour, relatively stable spending across customer groups, and a clear concentration of revenue in Clothing and Accessories.

A key finding is that subscribers do not spend significantly more than non-subscribers, highlighting an opportunity to convert more engaged repeat customers into subscribers.

The analysis also shows that discounts are common for several products, while highly rated products maintain consistently positive reviews. These insights can help guide future promotions, product selection, and customer targeting.

Overall, the project demonstrates a complete workflow from data cleaning in Python → SQL analysis in PostgreSQL → visualization in Power BI → business recommendations.